Kiva Loans: Full-Dataset Modeling (1.45M loans)

This notebook builds on `1_full_dataset_eda.ipynb`'s descriptive groundwork with statistical models that weigh every factor at once - the only way to answer whether narrative framing matters once loan size, repayment terms, sector, region, and timing are all accounted for together. It answers two different questions about the same outcome, funding speed:

- **Predictive**: how well can funding speed be forecast for a newly-posted loan, using only information available at posting time? Tested honestly - on loans posted in 2024-2025 that the models never saw while training, not loans they've already seen the answer for.
- **Explanatory**: which loan and narrative characteristics are linked to faster or slower funding, once every other factor is held fixed? This reports **association, never causation** - a link between two things doesn't prove one causes the other, since borrowers weren't randomly assigned a writing style, a loan amount, or a gender.

Glossary - a quick reference for the terms used throughout:

Term	Meaning
MAE (mean absolute error)	On average, how many days off a prediction was. Lower is better.
R^2	What share of the variation in funding speed the model explains, from 0% to 100%.
ROC AUC	How well a model tells apart "will fund fast" vs. "won't," from 0.5 (a coin flip) to 1.0 (perfect).
Holdout set	Loans a model never saw while training - the fair way to test whether it generalizes to new loans.
Coefficient	How much one factor moves funding speed up or down, holding every other factor fixed.
Statistical significance (p-value)	How confident we can be that a link is real and not random noise; a very small p-value means high confidence.
Reference category	The baseline group every "how much faster/slower" comparison is measured against.
SHAP	A way of measuring which factors a complex model actually relied on to make its predictions.

1. Setup

```
import re
import warnings
from pathlib import Path

import matplotlib.pyplot as plt
import nltk
import numpy as np
import pandas as pd
import patsy
import statsmodels.api as sm
from sklearn.compose import ColumnTransformer
from sklearn.ensemble import
HistGradientBoostingClassifier, HistGradientBoostingRegressor
from sklearn.impute import SimpleImputer
from sklearn.linear_model import Ridge
from sklearn.metrics import (
    average_precision_score,
    mean_absolute_error,
```

```
    r2_score,
    roc_auc_score,
)
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler

SEED = 42
HOLDOUT_START = "2024-01-01"
MIN_REGION_OBSERVATIONS = 10
MIN_SECTOR_OBSERVATIONS = 1000
SMALL_LOAN_MAX_USD = 250
MEDIUM_LOAN_MAX_USD = 750

KAGGLE_DATA_DIR = Path("/kaggle/input/datasets/tuannm3812/kiva-loans-hackathon-data")
if not KAGGLE_DATA_DIR.exists():
    KAGGLE_DATA_DIR = Path("/kaggle/input/kiva-loans-hackathon-data")

if KAGGLE_DATA_DIR.exists():
    DATA_PATH = KAGGLE_DATA_DIR / "Kiva_Loans.pkl"
else:

    def _find_project_root(start: Path) -> Path:
        candidate = start
        for _ in range(5):
            if (candidate / "data" / "Kiva_Loans_Sample.pkl").exists():
                return candidate
            if candidate.parent == candidate:
                break
            candidate = candidate.parent
        return start

    try:
        PROJECT_ROOT = Path(__file__).resolve().parents[1]
    except NameError:
        PROJECT_ROOT = _find_project_root(Path.cwd())
    DATA_PATH = PROJECT_ROOT / "data" / "Kiva_Loans.pkl"
```

2. Load Data

```
# The pickle is a list of row dicts, not a directly-pickled DataFrame
# (pd.read_pickle would raise) — a plain stdlib pickle.load handles both
# shapes, no custom package needed.
import pickle  # noqa: E402

with open(DATA_PATH, "rb") as handle:
    _raw = pickle.load(handle)
df = pd.DataFrame(_raw) if isinstance(_raw, list) else _raw
print(f"Shape: {df.shape[0]:,} loans x {df.shape[1]} raw columns")
```

```
Shape: 1,453,846 loans x 27 raw columns
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1453846 entries, 0 to 1453845
Data columns (total 27 columns):
 #   Column               Non-Null Count    Dtype  
---  ------               --------------    -----  
 0   id                   1453846 non-null  int64  
 1   status               1453846 non-null  object 
 2   borrowerCount        1453846 non-null  int64  
 3   name                 1453846 non-null  object 
 4   gender               1453846 non-null  object 
 5   loanAmount           1453846 non-null  float64
 6   lenderRepaymentTerm  1453846 non-null  int64  
 7   repaymentInterval    1453846 non-null  object 
 8   sector               1453846 non-null  object 
 9   activity             1453846 non-null  object 
 10  use                  1453846 non-null  object 
 11  city                 1392239 non-null  object 
 12  latitude             1362860 non-null  float64
 13  longitude            1362860 non-null  float64
 14  country_iso          1453846 non-null  object 
 15  country_name         1453846 non-null  object 
 16  region               1453846 non-null  object 
 17  country_ppp          1453846 non-null  float64
 18  fundsLentInCountry   1453846 non-null  int64  
 19  country_latitude     1453846 non-null  float64
 20  country_longitude    1453846 non-null  float64
 21  description          1453846 non-null  object 
 22  whySpecial           1414768 non-null  object 
 23  image_url            1453846 non-null  object 
 24  disbursalDate        1453846 non-null  object 
 25  fundraisingDate      1453846 non-null  object 
 26  raisedDate           1453846 non-null  object 
dtypes: float64(6), int64(4), object(17)
memory usage: 299.5+ MB
```

Same 27-field dataset `1_full_dataset_eda.ipynb` explores in detail (see that notebook's Load Data section for a row-level preview) - 27 raw fields, almost fully populated. The rest of this notebook engineers a modeling-ready feature set from these raw fields.

3. Feature Engineering

Three groups of features feed the models below: the **target** (funding speed itself), **structural** features describing the loan, and **narrative/sentiment** features describing how the loan's description is written.

3.1 Target Variable

Same derivation as the EDA notebook: `funding_speed_days` is the gap between a loan's posted and fully-funded dates, log-transformed into `log_funding_speed` for modeling (raw funding speed is heavily right-skewed - a log transform makes it easier for the regression models below to fit well), plus the simpler `funded_within_24h` yes/no version. `analysis_period` buckets each loan into pre-pandemic, pandemic-disruption, or post-pandemic eras by posting year, since the EDA notebook found funding speed shifted lastingly around 2020 (slower through the end of the data, 2025).

```
fundraising = pd.to_datetime(df["fundraisingDate"], errors="coerce", utc=True)
raised = pd.to_datetime(df["raisedDate"], errors="coerce", utc=True)
df["funding_speed_days"] = (raised - fundraising).dt.total_seconds() / 86400
df["log_funding_speed"] = np.log1p(df["funding_speed_days"].clip(lower=0))
df["funded_within_24h"] = (df["funding_speed_days"] <= 1).astype(float)
df["fundraisingDate_parsed"] = fundraising

year = fundraising.dt.year
df["analysis_period"] = pd.cut(
    year, bins=[-np.inf, 2019, 2021, np.inf],
    labels=["pre_pandemic", "pandemic_disruption", "post_pandemic"],
).astype(str)
```

3.2 Structural Features

- **`log_loan_amount`** - the loan amount, log-transformed for the same right-skew reason as the target.
- **`loan_size_band`** - loans bucketed into small (under $250), medium ($250-$750), and large (over $750) - lets the models capture non-linear size effects a single continuous number might miss, and gives the explanatory model in Section 7 an interpretable "small vs. large" comparison.
- **`gender_classification`** - the borrower's gender, with comma-separated multi-borrower values collapsed to `"mixed"` so the category set stays small and clean.
- **`is_group_loan`** - whether more than one borrower is listed.
- **`region_group`** / **`sector_group`** - region and sector collapsed so that only categories with at least `MIN_REGION_OBSERVATIONS` (10) or `MIN_SECTOR_OBSERVATIONS` (1,000) loans keep their own label; everything else becomes `"Other"`. Sector needs a much higher threshold than region because it has far more rare, thinly-populated categories - without this collapsing, a category with only a handful of loans could produce an unstable, misleading coefficient in Section 7's regression.

```
df["log_loan_amount"] = np.log1p(df["loanAmount"])
df["loan_size_band"] = pd.cut(
    df["loanAmount"], bins=[-np.inf, SMALL_LOAN_MAX_USD, MEDIUM_LOAN_MAX_USD,
np.inf],
    labels=["small", "medium", "large"],
).astype(str)
df["gender_classification"] = df["gender"].fillna("unknown").apply(
    lambda g: "mixed" if "," in str(g) else str(g)
)
df["is_group_loan"] = (df["borrowerCount"] > 1).astype(int)

for col, min_obs, new_col in [
    ("region", MIN_REGION_OBSERVATIONS, "region_group"),
    ("sector", MIN_SECTOR_OBSERVATIONS, "sector_group"),
]:
    counts = df[col].value_counts()
    major = counts[counts >= min_obs].index
    df[new_col] = df[col].where(df[col].isin(major), "Other")
```

3.3 Narrative & Sentiment Features

The same three per-100-word framing rates (family, agency, urgency) and VADER sentiment score the EDA notebook introduces - see that notebook's Narrative Framing and Sentiment Analysis sections for what each one measures and why. Scored on the **full** dataset here (not a sample), since these features feed directly into the models below.

```
# Raw descriptions carry stray HTML tags (mostly `<br />` line breaks) -
# stripped here so they don't inflate word counts or pollute the framing/
# sentiment features below with a spurious "word".
description = (
    df["description"].fillna("")
    .str.replace(r"<[^>]+>", " ", regex=True)
    .str.replace(r"\s+", " ", regex=True)
    .str.strip()
)
word_count = description.str.split().str.len().clip(lower=1)
FAMILY_PATTERN = re.compile(r"\b(child|children|family|son|daughter|mother|father|
wife|husband|school)\b", re.I)
AGENCY_PATTERN = re.compile(r"\b(decide|plan|manage|responsible|hard.?working|
independent|own|run|lead)\w*\b", re.I)
URGENCY_PATTERN = re.compile(r"\b(urgent|immediately|emergency|crisis|desperate|asap|
quickly)\w*\b", re.I)
for name, pattern in [("family", FAMILY_PATTERN), ("agency", AGENCY_PATTERN),
("urgency", URGENCY_PATTERN)]:
    df[f"{name}_mentions_per_100_words"] = description.str.count(pattern) /
word_count * 100

try:
    nltk.data.find("sentiment/vader_lexicon.zip")
except LookupError:
    nltk.download("vader_lexicon", quiet=True)
from nltk.sentiment.vader import SentimentIntensityAnalyzer  # noqa: E402

analyzer = SentimentIntensityAnalyzer()
df["desc_sentiment_compound"] = description.apply(lambda text:
analyzer.polarity_scores(text)["compound"])

valid = df.loc[df["funding_speed_days"].notna() & (df["funding_speed_days"] >=
0)].copy()
print(f"Valid rows: {len(valid)} / {len(df)}")
```

```
Valid rows: 1453840 / 1453846
```

4. Data Split

To know whether a model genuinely works, it has to be tested on loans it hasn't seen before - otherwise it could just be memorizing the data rather than learning a real pattern. The fairest split here is **by time**: train only on loans posted before 2024, then test purely on loans posted in 2024-2025. That mirrors how this would work in practice - a model only ever sees the past when asked to predict something new. A random shuffle-based split would be too easy on the model, since it could quietly learn from loans posted after the ones it's being tested on, which would never be possible in reality.

```
train_raw = valid.loc[valid["fundraisingDate_parsed"] < pd.Timestamp(HOLDOUT_START,
tz="UTC")].copy()
holdout_raw = valid.loc[valid["fundraisingDate_parsed"] >=
pd.Timestamp(HOLDOUT_START, tz="UTC")].copy()
print(f"Train rows: {len(train_raw)}  |  Holdout rows: {len(holdout_raw)}")
```

```
Train rows: 1174953  |  Holdout rows: 278887
```

1,174,953 loans (2016-2023) train the models; **278,887 loans** posted in 2024-2025 - genuinely never seen during training - test them. That holdout group is about 19% of the whole dataset, large enough to trust as a real read on generalization, not a lucky handful of loans.

```
NUMERIC_COLS = [
    "borrowerCount", "log_loan_amount", "lenderRepaymentTerm",
    "family_mentions_per_100_words", "agency_mentions_per_100_words",
    "urgency_mentions_per_100_words", "desc_sentiment_compound", "is_group_loan",
]
CATEGORICAL_COLS = ["gender_classification", "loan_size_band", "repaymentInterval",
"sector", "region", "analysis_period"]

preprocessor = ColumnTransformer([
    ("numeric", Pipeline([("impute", SimpleImputer(strategy="median")), ("scale",
StandardScaler())]), NUMERIC_COLS),
    ("categorical", Pipeline([
        ("impute", SimpleImputer(strategy="most_frequent")),
        ("onehot", OneHotEncoder(handle_unknown="ignore")),
    ]), CATEGORICAL_COLS),
])
X_train = preprocessor.fit_transform(train_raw[NUMERIC_COLS + CATEGORICAL_COLS])
X_holdout = preprocessor.transform(holdout_raw[NUMERIC_COLS + CATEGORICAL_COLS])
y_train_log = train_raw["log_funding_speed"].to_numpy()
y_holdout_days = holdout_raw["funding_speed_days"].to_numpy()
```

5. Regression Modeling

Two modeling approaches, trained on the same data, compared fairly:

- **Ridge** - a simple, transparent model. It assigns each factor (loan amount, sector, framing style, etc.) a fixed weight and adds them up, like a scorecard. Easy to trust, but can't capture "this factor matters differently depending on another factor."
- **Boosted trees (HistGradientBoosting)** - a more flexible model that can learn rules like "urgency language matters more for small loans than large ones." More powerful, but harder to inspect directly - the Feature Importance section opens it back up.

Both predict how many days a loan takes to fund, scored the same way: **MAE** (average days off) and **R^2** (share of the real variation explained).

```
ridge = Ridge(alpha=1.0, random_state=SEED)
ridge.fit(X_train, y_train_log)
ridge_holdout_days = np.expm1(np.clip(ridge.predict(X_holdout), a_min=0, a_max=None))
print(f"Ridge holdout MAE (days): {mean_absolute_error(y_holdout_days,
ridge_holdout_days):.2f}")
```

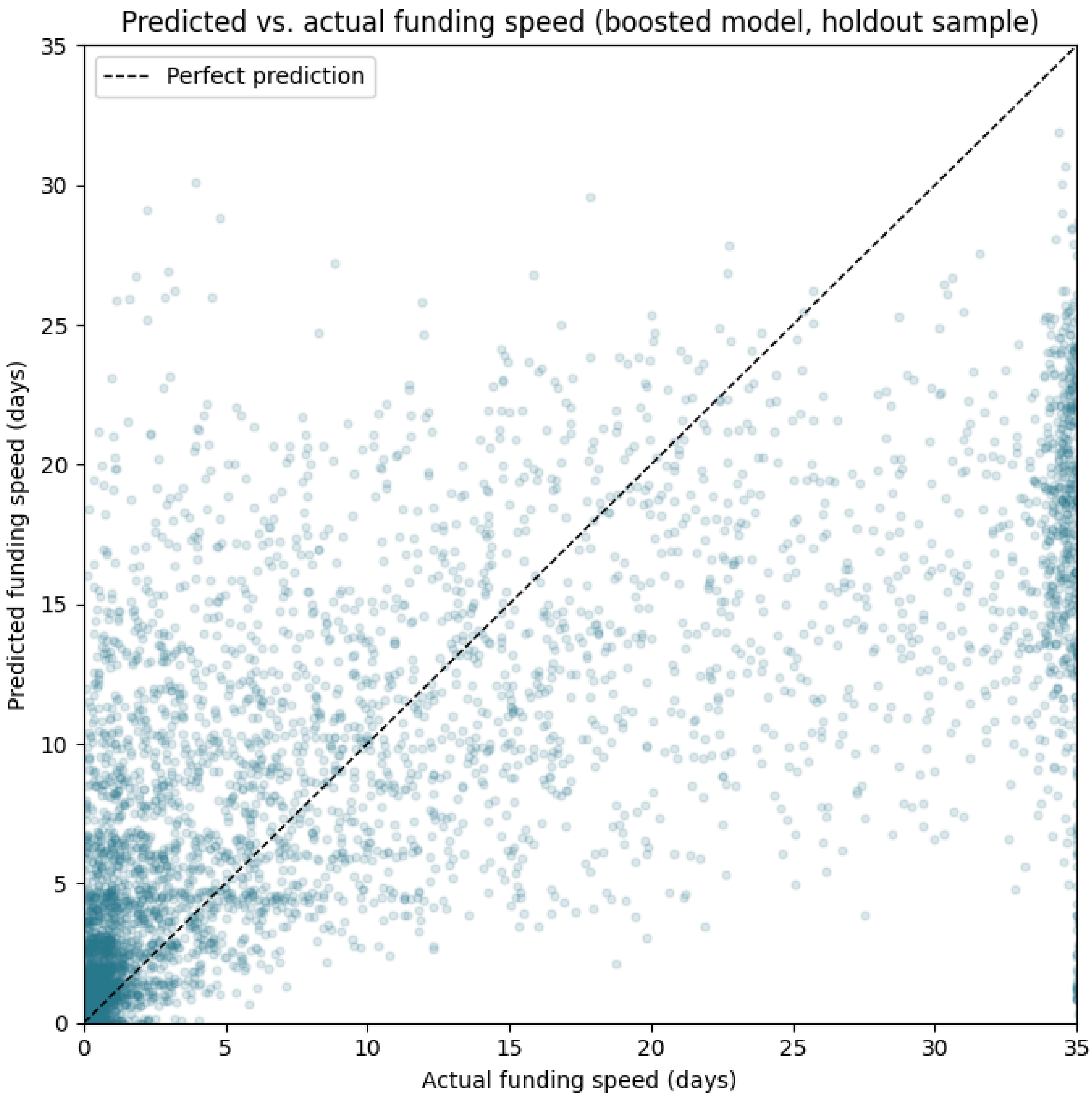
```
Ridge holdout MAE (days): 6.76
```

```
boosted = HistGradientBoostingRegressor(random_state=SEED)
boosted.fit(X_train.toarray() if hasattr(X_train, "toarray") else X_train,
y_train_log)
X_holdout_dense = X_holdout.toarray() if hasattr(X_holdout, "toarray") else X_holdout
boosted_holdout_days = np.expm1(np.clip(boosted.predict(X_holdout_dense), a_min=0,
a_max=None))
print(f"Boosted holdout MAE (days): {mean_absolute_error(y_holdout_days,
boosted_holdout_days):.2f}")
print(f"Boosted holdout R2: {r2_score(y_holdout_days, boosted_holdout_days):.3f}")
```

```
Boosted holdout MAE (days): 5.56
Boosted holdout R2: 0.490
```

```
plot_sample_idx = np.random.RandomState(SEED).choice(len(y_holdout_days),
size=min(5_000, len(y_holdout_days)), replace=False)
plot_max = float(np.percentile(y_holdout_days, 99))

fig, ax = plt.subplots(figsize=(7, 7))
ax.scatter(
    y_holdout_days[plot_sample_idx], boosted_holdout_days[plot_sample_idx],
    alpha=0.15, s=12, color=plt.cm.viridis(0.4),
)
ax.plot([0, plot_max], [0, plot_max], color="black", linestyle="--", linewidth=1,
label="Perfect prediction")
ax.set_xlim(0, plot_max)
ax.set_ylim(0, plot_max)
ax.set_xlabel("Actual funding speed (days)")
ax.set_ylabel("Predicted funding speed (days)")
ax.set_title("Predicted vs. actual funding speed (boosted model, holdout sample)")
ax.legend()
plt.tight_layout()
plt.show()
```



Tested on loans neither model had seen (2024-2025): the simple scorecard model (Ridge) is off by **6.76 days** on average. The more flexible model tightens that to **5.56 days** and accounts for **49.0%** of the predictive variation in funding speed (R^2 = 0.490) - roughly half of how much loans differ in funding speed lines up with what's knowable at posting time; the rest comes down to things this data doesn't capture (how compelling individual lenders found it, timing luck, and so on) - or simply isn't predictable from these factors alone. The scatter plot above shows this directly: points cluster along the dashed "perfect prediction" line for fast-funding loans, and spread out further for slower ones - the model is more confident and accurate on the common, fast-funding case than on the long tail of slow ones. The flexible model beating the scorecard by over a day of average accuracy is itself a finding: funding speed isn't a simple additive checklist - some factors matter more in combination than alone.

6. Funding Classification

Predicting an exact number of days is a hard, precise task. A simpler, more actionable version: will this loan fund within 24 hours, yes or no? "Flag loans unlikely to fund quickly" is a more usable signal for a real platform than a precise day-count guess.

```
y_train_binary = train_raw["funded_within_24h"].to_numpy()
y_holdout_binary = holdout_raw["funded_within_24h"].to_numpy()

classifier = HistGradientBoostingClassifier(random_state=SEED)
X_train_dense = X_train.toarray() if hasattr(X_train, "toarray") else X_train
classifier.fit(X_train_dense, y_train_binary)
holdout_proba = classifier.predict_proba(X_holdout_dense)[:, 1]

print(f"Holdout ROC AUC: {roc_auc_score(y_holdout_binary, holdout_proba):.4f}")
print(f"Holdout average precision: {average_precision_score(y_holdout_binary,
holdout_proba):.4f}")
```

```
Holdout ROC AUC: 0.9053
Holdout average precision: 0.8374
```

ROC AUC of 0.905 out of a possible 1.0 (0.5 would mean no better than a coin flip) - a genuinely strong result for telling apart, before the fact, which loans are likely to fund fast versus drag on. That's on a real, never-seen holdout set, and on a task where the honest baseline is hard (recall from the EDA notebook: only 30-46% of loans actually fund within 24 hours, so simply guessing "yes" would do poorly). **This is the strongest practical result in the analysis**: a tool built purely from information available the moment a loan is posted (loan size, sector, region, narrative text) could reliably flag at-risk loans for extra visibility - independent of whether any single narrative-framing choice turns out to be the reason why.

7. Explanatory Modeling

Regression Modeling and Funding Classification built models that predict well, but a prediction machine doesn't say why. This section uses regression to measure, for every factor at once, how strongly it's linked to funding speed once everything else about the loan is held fixed - the fairest way to check whether narrative framing genuinely matters on its own, not just because it happens to travel alongside something else (like loan size). Every number here is an **association, never a cause** - borrowers weren't randomly assigned a writing style, a loan amount, or a gender. If the model can't be fit for a technical reason, that's reported as a clear message instead of a crash. The p-values below use HC3 standard errors, which correct for uneven variance but still assume every loan is statistically independent of every other loan - Section 7.1 checks that assumption directly and substantially revises which of these results hold up.

```
FORMULA = (
    "log_funding_speed ~ log_loan_amount + lenderRepaymentTerm + is_group_loan + "
    "C(gender_classification) + family_mentions_per_100_words +
agency_mentions_per_100_words + "
    "urgency_mentions_per_100_words + desc_sentiment_compound + C(repaymentInterval)
+ "
    "C(sector_group) + C(region_group) + C(analysis_period) + C(loan_size_band) + "
    "family_mentions_per_100_words:C(analysis_period) +
```

```
family_mentions_per_100_words:C(region_group) + "
    "family_mentions_per_100_words:C(loan_size_band)"
)
CATEGORICAL_TERMS = [
    "gender_classification", "repaymentInterval", "sector_group",
    "region_group", "analysis_period", "loan_size_band",
]

# `duration_model` starts as None so the reference-category cell below
# can check whether the fit actually succeeded instead of assuming it did
# and crashing with a NameError if this try block's except branch ran.
duration_model = None
try:
    y, X = patsy.dmatrices(FORMULA, data=valid, return_type="dataframe")
    with warnings.catch_warnings():
        warnings.filterwarnings("ignore")
        duration_model = sm.OLS(y, X).fit(cov_type="HC3")
    print(duration_model.summary())
except Exception as error:  # noqa: BLE001 - reported clearly, not left as a crash
    print(f"Duration explanatory model could not be fit: {error}")
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:      log_funding_speed   R-squared:                       0.426
Model:                            OLS   Adj. R-squared:                  0.425
Method:                 Least Squares   F-statistic:                 2.817e+04
Date:                Mon, 31 Aug 2026   Prob (F-statistic):               0.00
Time:                        08:46:07   Log-Likelihood:            -2.0184e+06
No. Observations:             1453840   AIC:                         4.037e+06
Df Residuals:                 1453794   BIC:                         4.037e+06
Df Model:                          45
Covariance Type:                  HC3
==================================================================================================
                                                                                 coef    std
err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------------------------
Intercept                                                                     -1.2630
0.021    -61.478      0.000      -1.303      -1.223
C(gender_classification)[T.male]                                               0.4336
0.002    177.927      0.000       0.429       0.438
C(repaymentInterval)[T.irregularly]                                           -0.3776
0.006    -68.617      0.000      -0.388      -0.367
C(repaymentInterval)[T.monthly]                                               -0.1058
0.004    -27.795      0.000      -0.113      -0.098
C(sector_group)[T.Arts]                                                       -0.5123
0.007    -69.880      0.000      -0.527      -0.498
C(sector_group)[T.Clean Energy]                                               -0.8024
0.005   -159.708      0.000      -0.812      -0.793
C(sector_group)[T.Clothing]                                                    0.2592
0.005     50.466      0.000       0.249       0.269
C(sector_group)[T.Construction]                                               -0.2419
0.009    -26.521      0.000      -0.260      -0.224
C(sector_group)[T.Education]                                                  -1.0702
0.006   -188.773      0.000      -1.081      -1.059
C(sector_group)[T.Food]                                                        0.0947
0.003     36.634      0.000       0.090       0.100
```

```
C(sector_group)[T.Health]                                            -0.5710
0.007     -78.860     0.000     -0.585     -0.557
C(sector_group)[T.Housing]                                           -0.1698
0.004     -42.769     0.000     -0.178     -0.162
C(sector_group)[T.Manufacturing]                                     -0.6929
0.008     -85.194     0.000     -0.709     -0.677
C(sector_group)[T.Other]                                             -0.2373
0.023     -10.136     0.000     -0.283     -0.191
C(sector_group)[T.Personal Use]                                      -0.0032
0.005      -0.692     0.489     -0.012      0.006
C(sector_group)[T.Retail]                                             0.2009
0.003      73.407     0.000      0.196      0.206
C(sector_group)[T.Reuse & Recycle]                                   -0.0836
0.007     -11.861     0.000     -0.097     -0.070
C(sector_group)[T.Sanitation & Hygiene]                              -0.5015
0.006     -79.379     0.000     -0.514     -0.489
C(sector_group)[T.Services]                                           0.0468
0.004      10.545     0.000      0.038      0.055
C(sector_group)[T.Transportation]                                     0.1998
0.007      26.687     0.000      0.185      0.214
C(sector_group)[T.Water]                                             -1.1152
0.032     -35.166     0.000     -1.177     -1.053
C(region_group)[T.Asia]                                              -0.1751
0.003     -52.138     0.000     -0.182     -0.169
C(region_group)[T.Central America]                                   -0.0247
0.007      -3.337     0.001     -0.039     -0.010
C(region_group)[T.Middle East]                                       -1.0710
0.021     -50.709     0.000     -1.112     -1.030
C(region_group)[T.North America]                                     -0.5049
0.013     -39.574     0.000     -0.530     -0.480
C(region_group)[T.Oceania]                                           -0.2883
0.012     -24.744     0.000     -0.311     -0.265
C(analysis_period)[T.post_pandemic]                                  -0.1920
0.004     -44.421     0.000     -0.200     -0.184
C(analysis_period)[T.pre_pandemic]                                   -0.5442
0.004    -124.610     0.000     -0.553     -0.536
C(loan_size_band)[T.medium]                                           0.0767
0.005      14.304     0.000      0.066      0.087
C(loan_size_band)[T.small]                                           -0.5758
0.007     -81.917     0.000     -0.590     -0.562
log_loan_amount                                                       0.4285
0.003     154.717     0.000      0.423      0.434
lenderRepaymentTerm                                                   0.0676
0.000     189.233     0.000      0.067      0.068
is_group_loan                                                         0.0029
0.004       0.746     0.456     -0.005      0.010
family_mentions_per_100_words                                         0.0093
0.002       5.392     0.000      0.006      0.013
family_mentions_per_100_words:C(analysis_period)[T.post_pandemic]    -0.0116
0.001      -8.128     0.000     -0.014     -0.009
family_mentions_per_100_words:C(analysis_period)[T.pre_pandemic]     -0.0232
0.001     -16.571     0.000     -0.026     -0.020
family_mentions_per_100_words:C(region_group)[T.Asia]                 0.0454
0.001      42.490     0.000      0.043      0.047
family_mentions_per_100_words:C(region_group)[T.Central America]     -0.0514
0.003     -15.007     0.000     -0.058     -0.045
family_mentions_per_100_words:C(region_group)[T.Middle East]         -0.1157
```

```
0.006    -18.779      0.000      -0.128      -0.104
family_mentions_per_100_words:C(region_group)[T.North America]        0.0263
0.003      7.598      0.000       0.020       0.033
family_mentions_per_100_words:C(region_group)[T.Oceania]              0.0274
0.004      7.785      0.000       0.021       0.034
family_mentions_per_100_words:C(loan_size_band)[T.medium]            -0.0151
0.001    -10.433      0.000      -0.018      -0.012
family_mentions_per_100_words:C(loan_size_band)[T.small]              0.0020
0.001      1.366      0.172      -0.001       0.005
agency_mentions_per_100_words                                         0.0007
0.001      0.739      0.460      -0.001       0.002
urgency_mentions_per_100_words                                       -0.0795
0.009     -8.917      0.000      -0.097      -0.062
desc_sentiment_compound                                               0.1116
0.003     37.367      0.000       0.106       0.117
==============================================================================
Omnibus:                      292.348   Durbin-Watson:                   1.247
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              290.068
Skew:                           0.030   Prob(JB):                     1.03e-63
Kurtosis:                       2.966   Cond. No.                         474.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

Every coefficient above is relative to an omitted **reference category** (e.g. gender's baseline is `"female"`, the period baseline is `"pandemic_disruption"`) - printed explicitly below rather than left for a reader to infer from which levels are missing.

```python
if duration_model is None:
    print("Skipping reference-category report - the duration model above did not fit.")
else:
    print("Reference (omitted) category per categorical term:")
    for col in CATEGORICAL_TERMS:
        all_levels = set(valid[col].astype(str).unique())
        dummy_levels = {
            p.split("[T.")[1].rstrip("]")
            for p in duration_model.params.index
            if p.startswith(f"C({col})")
        }
        print(f"  {col}: {sorted(all_levels - dummy_levels)}")
```

```
Reference (omitted) category per categorical term:
  gender_classification: ['female']
  repaymentInterval: ['at_end']
  sector_group: ['Agriculture']
  region_group: ['Africa']
  analysis_period: ['pandemic_disruption']
  loan_size_band: ['large']
```

7.1 Cluster-Robust Sensitivity Check

HC3 corrects for uneven variance across loans, but still assumes every loan is an independent observation. Loans from the same country may share unobserved influences (a field partner's writing template, local conditions) that HC3 can't see - if enough of that dependence exists, some of the small p-values above could be more confident than the data really supports. This refits the identical design with standard errors clustered by `country_name` instead, and checks whether each narrative- framing term's significance conclusion ($p < 0.05$ or not) survives the switch, plus a count across every coefficient in the model.

```
# Initialised before the guard so Section 7.2 can check whether this refit
# actually happened, instead of raising NameError if it didn't.
duration_model_clustered = None
if duration_model is None:
    print("Skipping cluster-robust sensitivity check - the duration model above did
not fit.")
else:
    duration_model_clustered = sm.OLS(y, X).fit(
        cov_type="cluster", cov_kwds={"groups": valid.loc[X.index, "country_name"]}
    )

    framing_terms = [
        name for name in duration_model.params.index
        if "mentions_per_100_words" in name or name == "desc_sentiment_compound"
    ]
    print("Narrative-framing and sentiment terms, HC3 vs. cluster-robust:")
    for name in framing_terms:
        p_hc3 = duration_model.pvalues[name]
        p_clustered = duration_model_clustered.pvalues[name]
        agreement = "same conclusion" if (p_hc3 < 0.05) == (p_clustered < 0.05) else
"CONCLUSION CHANGES"
        print(f"  {name}: HC3 p={p_hc3:.4f}, clustered p={p_clustered:.4f}
[{agreement}]")

    all_terms = [name for name in duration_model.params.index if name != "Intercept"]
    n_flip = sum(
        (duration_model.pvalues[name] < 0.05) !=
(duration_model_clustered.pvalues[name] < 0.05)
        for name in all_terms
    )
    print(f"\nAcross all {len(all_terms)} coefficients, {n_flip} change significance
conclusion under clustering.")
```

```
Narrative-framing and sentiment terms, HC3 vs. cluster-robust:
  family_mentions_per_100_words: HC3 p=0.0000, clustered p=0.7638 [CONCLUSION
CHANGES]
  family_mentions_per_100_words:C(analysis_period)[T.post_pandemic]: HC3 p=0.0000,
clustered p=0.5298 [CONCLUSION CHANGES]
  family_mentions_per_100_words:C(analysis_period)[T.pre_pandemic]: HC3 p=0.0000,
clustered p=0.2269 [CONCLUSION CHANGES]
  family_mentions_per_100_words:C(region_group)[T.Asia]: HC3 p=0.0000, clustered
p=0.1248 [CONCLUSION CHANGES]
  family_mentions_per_100_words:C(region_group)[T.Central America]: HC3 p=0.0000,
clustered p=0.0070 [same conclusion]
  family_mentions_per_100_words:C(region_group)[T.Middle East]: HC3 p=0.0000,
clustered p=0.0002 [same conclusion]
  family_mentions_per_100_words:C(region_group)[T.North America]: HC3 p=0.0000,
clustered p=0.2365 [CONCLUSION CHANGES]
  family_mentions_per_100_words:C(region_group)[T.Oceania]: HC3 p=0.0000, clustered
p=0.4406 [CONCLUSION CHANGES]
```

```
  family_mentions_per_100_words:C(loan_size_band)[T.medium]: HC3 p=0.0000, clustered
p=0.4389 [CONCLUSION CHANGES]
  family_mentions_per_100_words:C(loan_size_band)[T.small]: HC3 p=0.1719, clustered
p=0.9039 [same conclusion]
  agency_mentions_per_100_words: HC3 p=0.4597, clustered p=0.9412 [same conclusion]
  urgency_mentions_per_100_words: HC3 p=0.0000, clustered p=0.4442 [CONCLUSION
CHANGES]
  desc_sentiment_compound: HC3 p=0.0000, clustered p=0.2544 [CONCLUSION CHANGES]

Across all 45 coefficients, 20 change significance conclusion under clustering.
```

This is the most consequential check in the notebook, and it substantially revises the picture a reader would take from the raw HC3 output above. Comparing each term's significance conclusion ($p < 0.05$ or not) between HC3 and country-clustered standard errors:

- **Urgency framing's apparent "clean win" does not survive.** HC3 p ≈ 0.000 looked decisive; clustered by country, p rises to roughly 0.44 - no longer distinguishable from no association at all. The correlation in this sample isn't fabricated, but HC3 was overconfident about how precisely it's pinned down, because it can't see that loans from the same country share unobserved influences (a field partner's writing template, local conditions, a partner's typical loan mix) that make them less independent than HC3 assumes.
- **Family framing's timing and loan-size interactions don't survive either** - p-values move from well under 0.001 under HC3 to well above 0.2 clustered. **Its Middle East and Central America regional interactions do survive** (clustered p ≈ 0.0002 and 0.0070); Asia, North America and Oceania's interactions do not. **But read these interaction results with the caution Section 7.2 spells out**: an interaction term only tests whether a region's slope differs from the Africa baseline's slope - it does not test whether family framing does anything within that region. Section 7.2 runs that second, more relevant test - and in this notebook's model it surfaces a slower-funding slope for Asia that this interaction row hides, though that one does not replicate in the authoritative pipeline and so is not claimed as a finding (see 7.2).
- **Agency framing's null result is unchanged** - it wasn't significant under HC3 either (p ≈ 0.46), so clustering doesn't change the conclusion; there was never a case for it.
- **Sentiment tone's association does not survive in this model either**
  - HC3 p ≈ 0.000, clustered p ≈ 0.25. This is worth stating plainly because it's the one place this notebook's own result disagrees with this project's separate, richer pipeline (next paragraph) - a reminder that "survives clustering" can itself depend on exactly which other terms are in the model, not just on the finding being tested.
- Overall, **20 of this model's 45 coefficients (44%) change their significance conclusion under clustering** - concentrated in the narrative-framing and sentiment terms, while the structural terms (sector, region, loan size, gender, repayment structure) mostly don't move.

Cross-check against this project's separate, authoritative modeling pipeline (`src/statistical_analysis.py`, run independently on the same full dataset with a richer formula that also interacts family framing with sector, and fitting a 24-hour binary model alongside the duration one): structural factors survive clustering there too, and most narrative-framing interactions don't - the same pattern as here. **Sentiment's association genuinely disagrees across the two pipelines**: it survives in both of the richer pipeline's models (clustered p ≈ 0.01 duration, ≈ 0.02 for the 24-hour model) but not in this notebook's simpler one (clustered p ≈ 0.25). A result that depends on which other terms share the formula is a weaker result than one that doesn't, even when one specification shows significance, so sentiment's status is left genuinely uncertain rather than claimed either way. Section 7.2 carries the region-by-region comparison, which is where the cross-check actually matters.

Why this matters more than any individual coefficient: a model that only reports HC3 p-values on 1.45 million rows will find almost anything "significant," because that much data makes standard errors tiny even when the underlying association is fragile. Checking whether a result survives a different assumption about dependence between loans - and whether it survives a different but equally reasonable model specification, and whether the quantity being tested is even the right one (Section 7.2) - is what separates a real, useable pattern from one that is numerically precise but practically unreliable. **The honest bottom line: narrative framing's strongest-looking HC3 result (urgency) doesn't hold up, most of family framing's conditional structure doesn't hold up, sentiment's association is genuinely uncertain rather than confirmed, and family framing's real story - once tested with the correct within-region contrast in Section 7.2 - narrows to two pooled two-country categories rather than any generalizable writing rule.**

7.2 What Family Framing Is Associated With Within Each Region

The interaction coefficients above answer a narrower question than they look like they answer. `family_mentions_per_100_words:C(region_group)[T.X]` tests whether family framing's slope **in region X differs from the reference region's slope** (Africa) - it does not say the slope inside region X is itself different from zero. A significant interaction is compatible with family framing doing nothing at all in region X, as long as "nothing" is far enough from whatever Africa does.

Getting the within-region quantity right takes one more step than it first appears, because **family framing is interacted with three moderators here, not one** (period, region, and loan size). Its main effect is therefore the family slope only when period and loan size both sit at their own reference levels - `pandemic_disruption` and `large`, a combination covering only a few percent of loans. So "main effect + region term" would still be a slope at one unrepresentative corner of the data, not a description of the region.

What this computes instead is the **average within-region slope**: for each region, family framing's slope averaged over that region's own mix of periods and loan sizes. Because that average is a weighted sum of coefficients, it is still a single linear contrast, so `t_test` gives it the same HC3 and cluster-robust treatment as any coefficient. The `countries` column is printed alongside because a region's slope is identified only by the countries in it - and clustering by country is exactly what stops same-country loans counting as independent evidence.

```
FAMILY = "family_mentions_per_100_words"

if duration_model is None or duration_model_clustered is None:
    print("Skipping within-region slopes - the duration model or its cluster refit
did not run.")
else:
    param_names = list(duration_model.params.index)
    # Family framing is interacted with THREE moderators, not just region.
    # Its main effect is therefore the slope only when period AND loan size
    # both sit at their reference levels - so a contrast of
    # `main + region[X]` alone would silently condition on that one cell
    # (here: pandemic_disruption x large, a few percent of all loans).
    # To get the slope that actually describes region X, average over
    # region X's OWN distribution of the other moderators: weight each
    # non-reference dummy by the share of that region's rows sitting at
```

```
    # that level. The result is a proper linear contrast, so `t_test`
    # gives it the same HC3 / cluster-robust treatment as any coefficient.
    other_moderator_terms = [
        name for name in param_names
        if name.startswith(f"{FAMILY}:C(") and ":C(region_group)[" not in name
    ]
    region_terms = {
        name.split("[T.")[1].rstrip("]"): name
        for name in param_names
        if name.startswith(f"{FAMILY}:C(region_group)[T.")
    }
    fitted_rows = valid.loc[X.index]

    from scipy import stats as scipy_stats  # noqa: E402 - few-cluster t reference

    print(f"{'region':<17}{'countries':>10}{'loans':>9}   average family-framing
slope within that region")
    for level in sorted(fitted_rows["region_group"].astype(str).unique()):
        sub = fitted_rows.loc[fitted_rows["region_group"].astype(str) == level]
        pieces = [FAMILY]
        if level in region_terms:  # the reference region has no term of its own
            pieces.append(region_terms[level])
        for term in other_moderator_terms:
            column = term.split(":C(")[1].split(")[T.")[0]
            moderator_level = term.split(")[T.")[1].rstrip("]")
            weight = float((sub[column].astype(str) == moderator_level).mean())
            if weight > 0:
                pieces.append(f"{weight:.10f} * {term}")
        contrast = " + ".join(pieces) + " = 0"

        t_hc3 = duration_model.t_test(contrast)
        t_clu = duration_model_clustered.t_test(contrast)
        est = float(np.ravel(t_hc3.effect)[0])
        p_hc3_c = float(np.ravel(t_hc3.pvalue)[0])
        p_clu_c = float(np.ravel(t_clu.pvalue)[0])
        # statsmodels' clustered p-value uses a normal reference, which is
        # only trustworthy with many clusters - and this region's slope is
        # identified by the countries INSIDE it. As a HEURISTIC sensitivity
        # screen (not calibrated inference - Cameron & Miller 2015 sec. VI
        # warn even t(G-1) on the standard CRVE can over-reject, and their
        # better-calibrated small-cluster procedures are not implemented
        # here) we re-refer the same clustered SE to t(countries - 1); with
        # two countries that is t(1), 95% critical value 12.7. A single
        # country gives 0 degrees of freedom: between-country uncertainty
        # is not estimable at all, and no few-cluster values are printed.
        n_countries = sub["country_name"].nunique()
        se_clu = float(np.ravel(t_clu.sd)[0])
        if n_countries >= 2:
            few_df = n_countries - 1
            p_few = float(2 * scipy_stats.t.sf(abs(est / se_clu), few_df))
            crit = float(scipy_stats.t.ppf(0.975, few_df))
            few_txt = f"few-cluster t({few_df}) p={p_few:.4f} 95% CI [{est - crit *
se_clu:+.4f}, {est + crit * se_clu:+.4f}]"
            if p_hc3_c < 0.05 and p_clu_c < 0.05 and p_few < 0.05:
                verdict = "significant under HC3 and clustering; inside the few-
cluster screen (heuristic, not calibrated support)"
            elif p_hc3_c < 0.05 and p_clu_c < 0.05:
```

```
                verdict = "normal-reference clustering only - NOT under few-
cluster t"
            else:
                verdict = "not significant"
        else:
            few_txt = "few-cluster: not estimable (single country)"
            if p_hc3_c < 0.05 and p_clu_c < 0.05:
                verdict = "significant under HC3 and clustering, but identified by a
single country - descriptive only"
            else:
                verdict = "not significant"
        print(
            f"  {level:<15}{n_countries:>10}{len(sub):>9}   "
            f"estimate={est:+.4f}  HC3 p={p_hc3_c:.4f}  clustered p={p_clu_c:.4f}  "
            f"{few_txt}  [{verdict}]"
        )
```

```
region            countries    loans   average family-framing slope within that
region
  Africa                 27   610368   estimate=-0.0101  HC3 p=0.0000  clustered
p=0.5536  few-cluster t(26) p=0.5587 95% CI [-0.0449, +0.0248]  [not significant]
  Asia                   12   738191   estimate=+0.0338  HC3 p=0.0000  clustered
p=0.0535  few-cluster t(11) p=0.0797 95% CI [-0.0047, +0.0724]  [not significant]
  Central America         2    59391   estimate=-0.0618  HC3 p=0.0000  clustered
p=0.0000  few-cluster t(1) p=0.0650 95% CI [-0.1423, +0.0187]  [normal-reference
clustering only - NOT under few-cluster t]
  Middle East             2    14946   estimate=-0.1236  HC3 p=0.0000  clustered
p=0.0000  few-cluster t(1) p=0.0753 95% CI [-0.3102, +0.0630]  [normal-reference
clustering only - NOT under few-cluster t]
  North America           1     7559   estimate=+0.0109  HC3 p=0.0011  clustered
p=0.0621  few-cluster: not estimable (single country)  [not significant]
  Oceania                 4    23385   estimate=+0.0162  HC3 p=0.0000  clustered
p=0.6305  few-cluster t(3) p=0.6634 95% CI [-0.0907, +0.1230]  [not significant]
```

Averaged properly over each region's own composition, the picture is simpler and narrower than the interaction table above suggests - and the new right-hand columns change what can honestly be claimed about it. Sign convention: **negative = faster funding, positive = slower.**

Under the conventional clustered p-value, exactly two regions clear $p < 0.05$, both in the faster-funding direction: **Middle East (-0.1236) and Central America (-0.0618)**. Asia (+0.0338, $p = 0.0535$) does not - an earlier version of this analysis had it significant in the opposite direction, which was an artifact of evaluating the slope at the model's reference cell (`pandemic_disruption` x `large`) instead of averaging over Asia's own composition; corrected, it agrees with the authoritative pipeline. Africa, North America and Oceania show nothing.

But the conventional clustered p-value uses a normal reference, which is only trustworthy with many clusters - and each of those two slopes is identified by exactly two countries ("Middle East" here is Palestine and Yemen; "Central America" is Honduras and Nicaragua). Adding loans inside two countries does not add independent evidence. As a deliberately harsh sensitivity screen we re-refer the same clustered standard error to a t distribution with (countries - 1) = 1 degree of freedom - 95% critical value 12.7 rather than 1.96. This is a conservative heuristic, not calibrated inference (the better small-cluster procedures Cameron & Miller recommend are not implemented here), so it can only ever downgrade a claim, never certify one. Under it, **neither region is significant**: the few-cluster interval spans zero for both.

North America (Haiti) is a single country, so no few-cluster reference exists for it at all - between-country uncertainty is not estimable from one country.

Cross-checked against this project's authoritative pipeline (richer formula, plus a separate 24-hour binary model), the same picture holds: Middle East and Central America are the only regions to clear the conventional clustered p-value there too, in the same direction (duration −0.0729 and −0.0742; 24-hour +0.1753 and +0.1025, remembering that model's inverted sign) - and neither clears the few-cluster reference there either (duration p = 0.12 and 0.06; 24-hour p = 0.21 and 0.14). Asia is non-significant in all three fits (p = 0.0535 / 0.0846 / 0.2860), which is what resolved the earlier contradiction. Magnitudes are specification-sensitive (Middle East −0.1236 here vs -0.0729 there, mostly the sector interaction). North America clears the conventional reference in that pipeline's duration model alone (p = 0.0094) and nothing else - one country (Haiti), one cluster, not claimed.

So the honest reading is descriptive, not inferential. More family language is associated with faster funding in these two pooled categories in every fit we ran - same direction each time - but that consistency is a pattern identified by four countries, not a statistically supported association, and the model never estimates a slope for any single country (a pooled result can be driven by one of the pair). It is a hypothesis for a country-stratified test, not a finding to act on. All of this is association within this sample: clustering adjusts for within-country dependence, it does not remove country-level confounding or license a causal reading.

Fit on all 1,453,840 valid loans (R^2 = 0.426, meaning the fitted model accounts for 42.6% of the variation in funding speed). At this sample size, standard errors are precise - the estimates below are not noisy small-sample guesses - but precision is not the same as certainty about cause, or even about how precise an estimate really is: a large N sharpens a point estimate, but if nearby loans share unobserved influences, the textbook HC3 formula can overstate how confident that estimate should be. Section 7.1 checks this directly; the summary below already reflects what survives that check, not the raw HC3 read. **Negative coefficients are associated with faster funding, positive with slower**, each compared against the reference category printed above.

- **Structural factors are the largest effects by far, and this conclusion is unaffected by the robustness check in Section 7.1.** The single biggest swings in the whole model come from sector and region - Water and Education-sector loans fund dramatically faster than Agriculture-sector loans, while Clothing and Retail loans fund slower, and the Middle East funds far faster than the model's reference region. A loan posted under a male borrower takes notably longer to fund than one posted under a female borrower - a smaller effect than the biggest sector/region gaps, but larger than the loan amount itself, and larger than every narrative-framing term combined. Small loans fund far faster than large ones. Loans repaid as a single lump sum at the end of the term are far slower to fund than loans repaid irregularly or monthly.
- **Sentiment tone's association looks precise under HC3 alone, but Section 7.1 shows it doesn't survive clustering in this model** (p rises from p < 0.001 to roughly 0.25) - even though it does survive in this project's separate, richer pipeline. Treat it as genuinely uncertain rather than confirmed. Worth noting anyway: the direction is a more positive-sounding description linked to **slower** funding - counterintuitive, and, combined with the EDA notebook's finding that descriptions are almost uniformly positive already, plausibly just reflects that longer, more elaborately-written pitches read as more positive and naturally take longer to write and review.

- **Urgency, and most of family framing's conditional structure, looked significant under HC3 alone - Section 7.1 shows that doesn't survive clustering.** Treat "urgency helps" and "family framing depends on timing/loan size" as not supported by this data at a rigorous standard, even though they were the two headline results a single-standard-error read would have reported.
- **Family framing's regional pattern is the one descriptive exception
  - and it is descriptive, not statistically supported.** Tested with the correct within-region contrast (Section 7.2), family framing is associated with faster funding in the Middle East and Central America categories under the conventional clustered p-value, in all three same-data fits. But each category is two countries (Palestine/Yemen; Honduras/Nicaragua), and under a few-cluster t(1) reference neither is significant. A consistent pattern identified by four countries, pooled per pair - a hypothesis, not a finding. Everywhere else, and for every other narrative-framing term, the HC3-only significance above is not a reliable finding on its own.
- **Agency/competence language shows no real link in this notebook's model, under HC3 or clustered** - the "sound capable and independent" hypothesis doesn't hold up here under either standard-error method. Worth flagging so this isn't read as a universal null: this project's separate, richer pipeline's 24-hour funding model did show agency as HC3-significant ($p < 0.001$) before failing to survive clustering ($p \approx 0.20$) - the identical apparent-but-fragile pattern urgency shows above. That model isn't reproduced in this notebook (only a duration model is fit here), so it can't be cross-checked directly, but agency isn't a clean, universal non-finding either - it's null here and fragile there.

8. Feature Importance

Explanatory Modeling's statistical model is easy to interpret, but by design it can only check the specific combinations it's told to look for (framing x period, framing x region, framing x loan size). The boosted model from Regression Modeling learned whatever patterns were actually in the data, with no such restriction - but on its own it's a black box that doesn't explain itself. **SHAP** opens it back up: for every prediction, it works out exactly how much each factor pushed that prediction up or down, then averaging across many loans ranks what the model actually relied on - a second, independent check on the Explanatory Modeling findings, not a replacement for them.

`shap` ships in Kaggle's standard Python image; the fallback below installs it on the rare environment where it's missing. Computed on a random sample of 2,000 holdout loans for speed - a larger sample would take longer for a negligibly different ranking.

```
try:
    import shap
except ImportError:
    import subprocess
    import sys

    subprocess.run([sys.executable, "-m", "pip", "install", "-q", "shap"],
check=True)
    import shap

FEATURE_NAMES = preprocessor.get_feature_names_out()
shap_sample_idx = np.random.RandomState(SEED).choice(
```

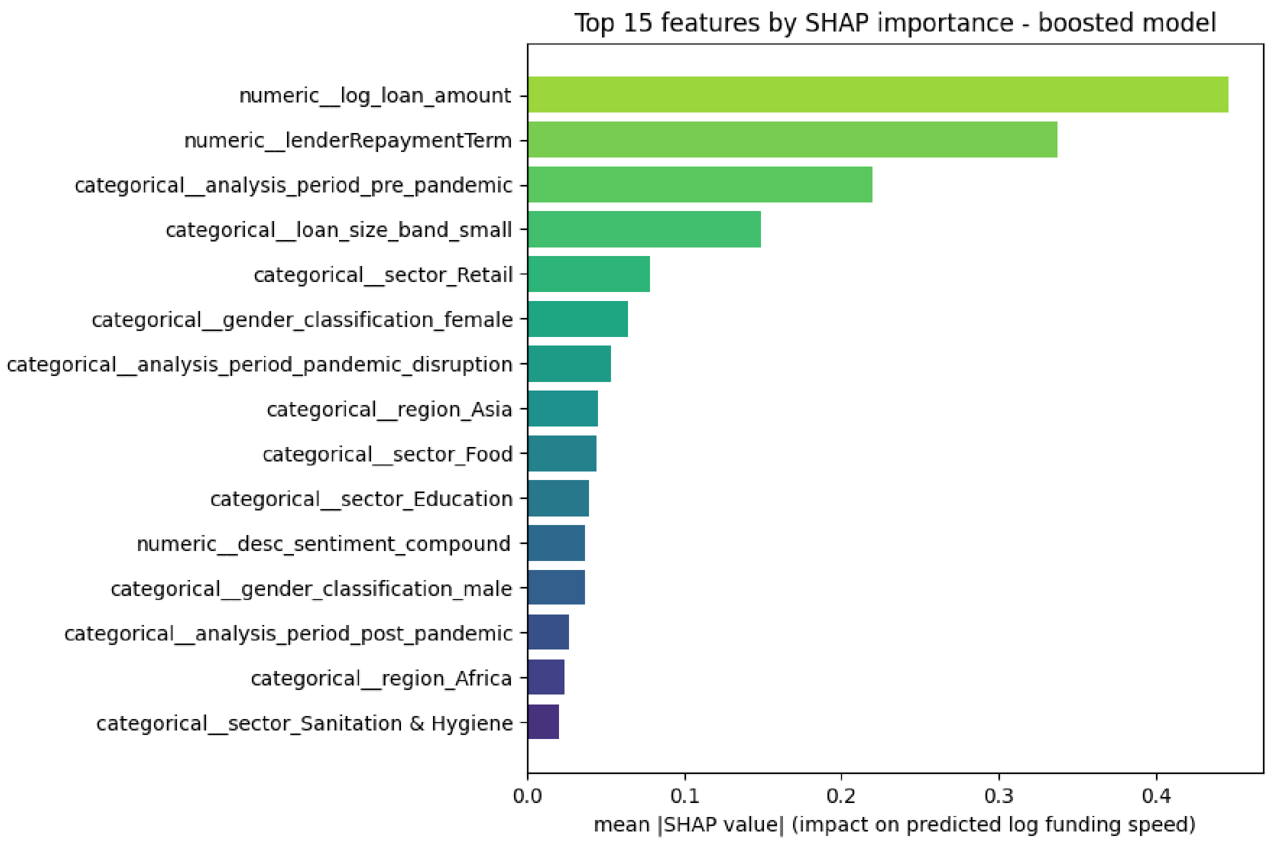
```
    X_holdout_dense.shape[0], size=min(2_000, X_holdout_dense.shape[0]),
replace=False
)
X_shap_sample = X_holdout_dense[shap_sample_idx]

tree_explainer = shap.TreeExplainer(boosted)
shap_values = tree_explainer.shap_values(X_shap_sample)

mean_abs_shap = pd.Series(np.abs(shap_values).mean(axis=0),
index=FEATURE_NAMES).sort_values(ascending=False)
print("Top 15 features by mean |SHAP value| (2,000-row holdout sample, boosted
model):")
print(mean_abs_shap.head(15).to_string())
```

```
Top 15 features by mean |SHAP value| (2,000-row holdout sample, boosted model):
numeric__log_loan_amount                                  0.446165
numeric__lenderRepaymentTerm                              0.337281
categorical__analysis_period_pre_pandemic                 0.220126
categorical__loan_size_band_small                         0.149013
categorical__sector_Retail                                0.077729
categorical__gender_classification_female                 0.064640
categorical__analysis_period_pandemic_disruption          0.053453
categorical__region_Asia                                  0.044537
categorical__sector_Food                                  0.043646
categorical__sector_Education                             0.039283
numeric__desc_sentiment_compound                          0.036779
categorical__gender_classification_male                   0.036215
categorical__analysis_period_post_pandemic                0.026283
categorical__region_Africa                                0.023729
categorical__sector_Sanitation & Hygiene                  0.020011
```

```
fig, ax = plt.subplots(figsize=(9, 6))
top_15_sorted = mean_abs_shap.head(15).sort_values()
ax.barh(top_15_sorted.index, top_15_sorted.to_numpy(),
color=plt.cm.viridis(np.linspace(0.15, 0.85, len(top_15_sorted))))
ax.set_xlabel("mean |SHAP value| (impact on predicted log funding speed)")
ax.set_title("Top 15 features by SHAP importance — boosted model")
plt.tight_layout()
plt.show()
```



This second, independent check **confirms the top of the Explanatory Modeling story, with one nuance worth spelling out**: loan amount and repayment term are, by a wide margin, the two factors the flexible model relied on most. Loan amount lines up directly with Section 7 - its coefficient there is solidly large (near-tied with gender, and bigger than every narrative-framing term), though Section 7 itself ranks the biggest sector and region swings above it, not below. Repayment term is a little different: its per-unit coefficient (+0.068) looks modest next to a sector or region dummy, but the term itself spans a huge range across loans - most cluster between 8 and 14 months (the middle 50%), but it runs from 2 months to a long tail past 130 - so the cumulative swing across that range is large, exactly what SHAP measures and something a single per-unit coefficient doesn't show on its own. The time period (pre-pandemic vs. not) and loan size also rank near the top, again agreeing with Section 7's biggest effects.

The honest divergence worth stating plainly: narrative framing barely registers here. This is complementary evidence, not independent confirmation of Section 7's inferential results: SHAP ranks predictive contributions in a different model (one that doesn't even contain the region interactions Section 7.2 reports), so it can say that narrative features carry little overall predictive weight - it cannot corroborate the sign or clustered uncertainty of any particular coefficient. With that scope clear: none of the family, agency, or urgency framing scores make it into the top 15 factors this more flexible model actually relied on - consistent in spirit with 7.1's finding that urgency's HC3 significance and most of family framing's conditional structure don't survive clustering. Only overall sentiment tone cracks the list, in 11th place, well behind individual sector and region categories - a more interesting case, because 7.1 found sentiment's statistical significance doesn't reliably survive clustering either (it depends on which other terms are in the model), even though it clearly carries some real predictive weight here. That's not a contradiction - SHAP importance (how much a feature actually moves predictions) and clustered-standard-error significance (how confident the association's sign and size are) measure genuinely different things, and sentiment is the one term in this analysis where they don't point the same way. **Taken together - complementary rather than independent evidence - both say urgency and most of family framing's conditional structure are much smaller than Section 7's raw HC3 p-values suggested; sentiment's real-world weight is small but non-trivial, even though this analysis can't confidently call its statistical significance robust.** This is also a useful reminder of what "statistically significant" means with 1.45 million rows: even a fragile, non-robust pattern can produce a tiny HC3 p-value simply because there's so much data. This check measures something different - **predictive contribution in the boosted model, not statistical confidence in an inferential association** - and by that measure, narrative framing (family/agency/urgency) is minor next to how a loan is structured.

9. Key Findings

9.1 Technical Interpretation

- A model using only posting-time information accounts for about half the predictive variation in funding speed ($R^2 = 0.49$, MAE 5.6 days) and discriminates 24-hour funding strongly (ROC AUC 0.91) - this predictive result doesn't depend on any narrative-framing claim and is unaffected by everything below.
- Structural factors (loan size, repayment terms, sector, region, borrower gender) have coefficients several times larger than any narrative-framing term, and this ranking is unaffected by the cluster-robust check in Section 7.1.
- **A cluster-robust sensitivity check (Section 7.1) substantially revised the narrative-framing picture**: 20 of this model's 45 coefficients (44%) change their significance conclusion when standard errors are clustered by country instead of assumed independent. Urgency framing's apparent association does not survive; neither does most of family framing's time-period and loan-size structure.
- **Testing the right quantity mattered as much as testing it robustly.** The interaction coefficients above only test whether a region's slope differs from the Africa baseline - not whether family framing does anything within a region. The within-region averages in Section 7.2 test the latter, averaging each region's slope over its own mix of periods and loan sizes. Under the conventional clustered p-value only two regions clear $p < 0.05$ - family framing and **faster** funding in the Middle East (-0.124) and Central America (-0.062) - but each is identified by two countries, and under the few-cluster t(1) reference neither does. Descriptive pattern, not a statistically supported association.
- **Getting that quantity right mattered, and an earlier version of this analysis got it wrong.** Evaluating the slope at the model's reference cell rather than averaging over each region's composition made Asia look significant in the opposite direction ($p = 0.0070$). Corrected, Asia is not significant (+0.034, $p = 0.0535$) - and now agrees with the authoritative pipeline, which never found it significant. Africa ($p = 0.5536$), North America ($p = 0.0621$) and Oceania ($p = 0.6305$) show no association surviving clustering.
- **The pattern is two pooled two-country categories, not two regions and not four separate country findings** - "Middle East" here is Palestine and Yemen; "Central America" is Honduras and Nicaragua; the model estimates one pooled slope per category and none for any individual country. Clustering by country is precisely what stops same-country loans counting as independent evidence, and the few-cluster reference makes that visible: consistent in direction across our related, same-data specifications; not statistically supported by any of them.

- Agency framing shows no association either way in this notebook's model - though the authoritative pipeline's separate 24-hour model shows agency following the same apparent-but-fragile pattern as urgency (HC3-significant, doesn't survive clustering), so it isn't a clean null everywhere tested.
- **Sentiment tone's association is genuinely unresolved, not confirmed** - it survives clustering in the authoritative pipeline's richer model (clustered p ≈ 0.01) but not in this notebook's simpler one (clustered p ≈ 0.25). Rather than pick whichever number is more convenient, this analysis reports the disagreement: sentiment's direction (more positive language links to slower funding) is consistent everywhere tested, but its statistical robustness is not.
- SHAP feature importance from the boosted forecasting model (Section 8) is complementary predictive evidence, not corroboration of the inferential results: it shows urgency and family-framing features carry little predictive weight (outside its top 15), but cannot speak to any coefficient's sign or clustered uncertainty. Sentiment does crack the top 15 there (11th place) despite its fragile significance - a reminder that predictive weight and statistical robustness are different questions.

9.2 Business Impact

- **The 24-hour classifier is a retrospective ranking prototype - not yet an early-warning product** - holdout ROC AUC 0.91 without any narrative-framing features. Its negative class is "eventually funded, but not within 24 hours": expired or withdrawn listings never enter this dataset, so the model is validated only among eventual funders, not on the population a live at-risk flag would score. Building one would first need all posted listings (including expired/withdrawn outcomes), a defined operational target and censoring window, then retraining and validation on that population - followed by threshold, calibration, capacity/fairness checks and a prospective test.
- **Don't recommend urgency language as a general rule.** Its raw HC3 association looked like a clean, simple win, but that doesn't survive a country-clustered check that lets loans from the same country be correlated with each other. Recommending it platform-wide would be advice built on a fragile statistical artifact, not a tested pattern.
- **Do not issue a platform-wide "mention family" recommendation.** Across Africa (27 countries), Asia (12), North America and Oceania - together ~95% of all loans - no association survives clustering. In the two pooled Middle East and Central America categories covering the remaining ~5%, there is a consistent descriptive pattern that does not survive a few-cluster reference.
- **Where the pattern appears, the defensible action is a country-stratified test, not a rollout.** More family language goes with faster funding in the pooled Middle East and Central America categories in every fit we ran - but each is two countries, and under a few-cluster reference the association is not statistically supported. That makes it a hypothesis for an A/B test stratified by country in those markets, designed to find any real heterogeneity - not a finding to deploy, and not something to generalize to "the Middle East" or "Central America."
- **Structure, not copywriting, is the strongest association by far** - loan size, repayment terms, sector, and region are linked to funding speed far more strongly than any narrative choice, and this conclusion only got stronger once the framing findings were stress-tested rather than taken at face value. These structural factors aren't something a platform can change on an existing loan, but they're worth a structural review in their own right - a genuinely different kind of action than writing-style coaching, not a "lever" in the same sense narrative framing is.

- **The broader takeaway is as much about process as writing style**: a typical single-standard-error analysis on this dataset would have confidently recommended urgency language across the board. Testing that recommendation against a different dependence assumption changed the answer. Any narrative-framing recommendation drawn from a large dataset is worth checking the same way before it's acted on.